

```
!pip install pandas numpy matplotlib seaborn plotly
```

Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages (5.24.1)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
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Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.1)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.12/dist-packages (from plotly) (9.1.4)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)

```
import pandas as pd

url = "https://static.openfoodfacts.org/data/en.openfoodfacts.org.products.csv.gz"

# Load only first 200,000 rows (safe size)
df = pd.read_csv(url, sep='\t', nrows=200000, low_memory=False)

df.head()
```

	code	url	creator	created_t	created_datetime	last_modified_t	last_modified_datetime	last_m...
0	54	http://world-en.openfoodfacts.org/product/0000...	kiliweb	1582569031	2020-02-24T18:30:31Z	1733085204	2024-12-01T20:33:24Z	
1	63	http://world-en.openfoodfacts.org/product/0000...	kiliweb	1673620307	2023-01-13T14:31:47Z	1750061386	2025-06-16T08:09:46Z	bc
2	114	http://world-en.openfoodfacts.org/product/0000...	kiliweb	1580066482	2020-01-26T19:21:22Z	1751035658	2025-06-27T14:47:38Z	
3	431	http://world-en.openfoodfacts.org/product/0000...	kiliweb	1714301712	2024-04-28T10:55:12Z	1714301721	2024-04-28T10:55:21Z	
4	105	http://world-en.openfoodfacts.org/product/0000...	kiliweb	1572117743	2019-10-26T19:22:23Z	1738073570	2025-01-28T14:12:50Z	

5 rows × 210 columns

```
df.shape

(200000, 210)
```

```
df.columns

Index(['code', 'url', 'creator', 'created_t', 'created_datetime',
      'last_modified_t', 'last_modified_datetime', 'last_modified_by',
      'last_updated_t', 'last_updated_datetime',
      ...,
      'choline_100g', 'phyllloquinone_100g', 'beta-glucan_100g',
      'inositol_100g', 'carnitine_100g', 'sulphate_100g', 'nitrate_100g',
      'acidity_100g', 'carbohydrates-total_100g', 'water_100g'],
      dtype='object', length=210)
```

```
df[['product_name', 'sugars_100g', 'proteins_100g', 'categories_tags']].head(10)
```

	product_name	sugars_100g	proteins_100g	categories_tags
0	Limonade artisanale a la rose	NaN	NaN	NaN
1	M&M white	NaN	NaN	NaN
2	Chocolate n3	NaN	NaN	NaN
3	Pâte de fruits	NaN	NaN	NaN
4	Paleta gran reserva - Sierra nevada-	NaN	NaN	en:beverages-and-beverages-preparations,en:bev...
5	NaN	NaN	NaN	NaN
6	NaN	NaN	NaN	NaN
7	NaN	NaN	NaN	NaN
8	Confiture extra citron de Menton	NaN	NaN	NaN
9	NaN	NaN	NaN	NaN

```
df = df[['product_name', 'sugars_100g', 'proteins_100g', 'categories_tags', 'ingredients_text']]
```

```
df = df.dropna(subset=['product_name', 'sugars_100g', 'proteins_100g'])
```

```
df = df[(df['sugars_100g'] >= 0) & (df['sugars_100g'] <= 100) &
(df['proteins_100g'] >= 0) & (df['proteins_100g'] <= 100)]
```

```
df.reset_index(drop=True, inplace=True)
```

```
def assign_category(tags):
    if pd.isna(tags):
        return "Other"

    tags = tags.lower()

    if "chocolate" in tags or "sweets" in tags:
        return "Sweets"
    elif "biscuits" in tags or "cookies" in tags:
        return "Biscuits"
    elif "beverages" in tags or "drinks" in tags:
        return "Beverages"
    elif "snacks" in tags:
        return "Snacks"
    elif "cereals" in tags or "breakfast" in tags:
        return "Cereals"
    else:
        return "Other"
```

```
df['primary_category'] = df['categories_tags'].apply(assign_category)
```

```
df['primary_category'].value_counts()
```

	count
primary_category	
Other	22478
Beverages	7645
Snacks	1311
Biscuits	930
Sweets	804
Cereals	79

dtype: int64

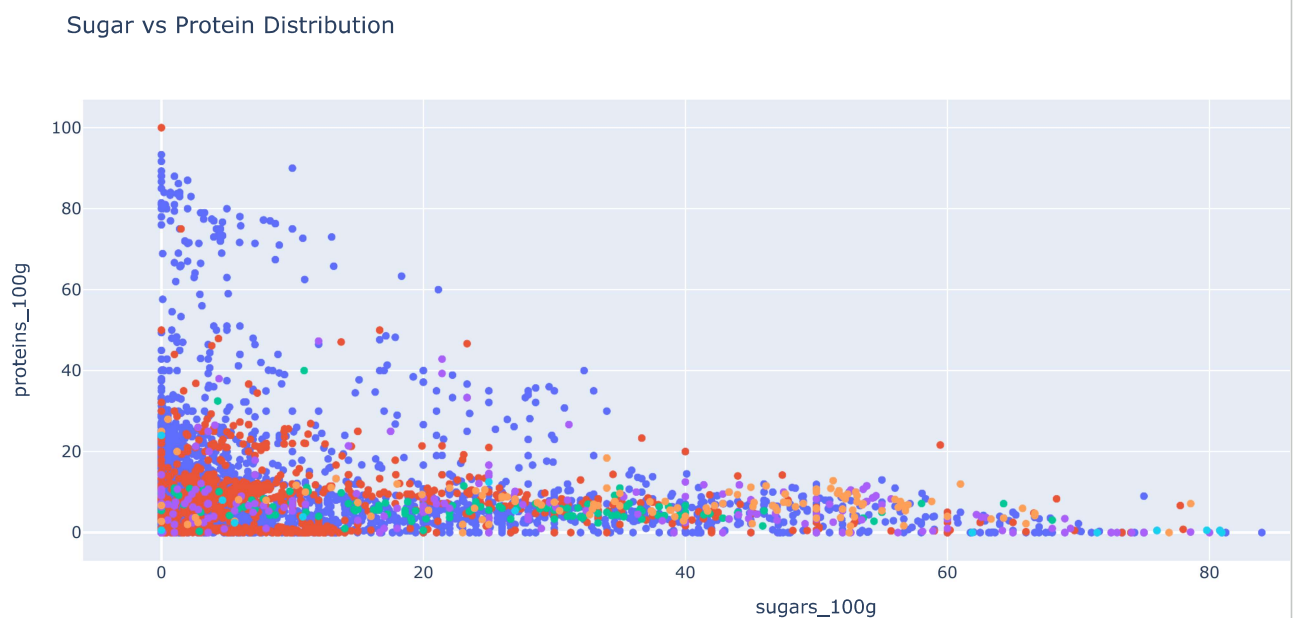
```
df[['sugars_100g', 'proteins_100g']].describe()
```

	sugars_100g	proteins_100g
count	33247.000000	33247.000000
mean	12.961551	9.721669
std	17.740572	12.737228
min	0.000000	0.000000
25%	1.333333	2.500000
50%	4.705882	6.670000
75%	17.699108	11.111111
max	100.000000	100.000000

```
import plotly.express as px

fig = px.scatter(
    df.sample(5000), # sample for performance
    x='sugars_100g',
    y='proteins_100g',
    color='primary_category',
    title='Sugar vs Protein Distribution'
)

fig.show()
```



```
gap_df = df[
    (df['proteins_100g'] > 10) &
    (df['sugars_100g'] < 5)
]
```

```
gap_df.shape
```

```
(6604, 6)
```

```
from collections import Counter

ingredients = gap_df['ingredients_text'].dropna().str.lower()

words = []
for item in ingredients:
    words.extend(item.split(' '))

common = Counter(words).most_common(10)
```

```
common
```

```
[(' salt', 1705),  
 (' water', 793),  
 (' sugar', 518),  
 (' niacin', 438),  
 (' yeast', 355),  
 (' dextrose', 284),  
 (' riboflavin', 267),  
 (' enzymes', 266),  
 (' spices', 259),  
 (' sea salt', 218)]
```

```
df.to_csv("cleaned_data.csv", index=False)
```

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